The cumulative distribution of Bernoulli's distribution is

$$F(x)=\begin{cases}0 & \text{if } x<0\\ q=1-p & \text{if } 0\le x<1\\ 1 & \text{if } x\ge 1\end{cases}$$

Mean :

$$E(X) = \sum_x x\, f(x) = 1\times p + 0\times(1-p) = p\,,$$

Note that, since X takes only the values 0 and 1, its expected value p is "never seen".

Variance :

$$V(X) = E(X^2)-\big(E(X)\big)^2 = \sum_x x^2 f(x) - p^2$$
$$= \left(1^2 p + 0^2 q\right) - p^2 = p(1-p) = pq \quad \text{where } q = 1-p$$

If X is a Bernoulli's random variable which follows Bernoulli distribution with parameter p, the mean μ and variance σ^2 are

$$\mu = p \qquad \text{and} \qquad \sigma^2 = pq$$

When $p = q = \frac{1}{2}$, the Bernoulli's distribution become

$$f(x)=\begin{cases}\frac{1}{2} & \text{for } x=0\\ \frac{1}{2} & \text{for } x=1\end{cases}$$

and the cumulative distribution is

$$F(x)=\begin{cases}0 & \text{if } x<0\\ \frac{1}{2} & \text{if } 0\le x<1\\ 1 & \text{if } x\ge 1\end{cases}$$

The mean and variance are respectively are $\frac{1}{2}$ and $\frac{1}{4}$

11.6.4 The Binomial Distribution

The Binomial Distribution is an important distribution which applies in some cases for repeated independent trials where there are only two possible outcomes: heads or tails, success or failure, defective item or good item, or many other such possible pairs. The probability of each outcome can be calculated using the multiplication rule, perhaps with a tree diagram.

Suppose a coin is tossed once. Let X denote the number of heads. Then $X \sim Ber(p)$, because we get either head $(X=1)$ or tail $(X=0)$ with probability p or $1-p$.

Suppose a coin is tossed n times. Let X denote the number of heads. Then X takes on the values 0, 1, 2, …, n. The probability for getting x number of heads is given by

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}\,,\ x = 0, 1, 2, ..., n.$$

12th_Maths_Vol 2_EM_CH 11_Probability Distributions_29-05-20.indd 213 16-12-2022 16:52:23